Mingen Li

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EDUCATION

University of California, Los Angeles Expected Dec 2028
M.S. in Computer Science
- Track: Artificial Intelligence
- Relevant Coursework: CS 180: **Algorithms and Complexity**, CS 163: **Deep Learning for Computer Vision**.

University of California, Berkeley Graduated May 2026
B.A. in Data Science
- Concentration: **Applied Mathematics and Modeling** | **Technical GPA** (CS, DS, Math & Stats)**: 3.967 / 4.0**
- Relevant Coursework: CS 61B: **Data Structures**, CS 188: **Artificial Intelligence**, CS 189: **Machine Learning.**

INTERNSHIP EXPERIENCE

AI Engineer – **Trial Data** Jun 2026 – Aug 2026
- Built an **agentic clinical-trial data insight system** using **planner-based task decomposition**, **schema-aware RAG**, and **LLM-based validation** to assess **treatment** and **visit adherence**, identify risks, and prioritize clinical research coordinator (CRC) follow-up.
- Evaluated the system against a non-agentic baseline on **50 clinical-trial cases**, increasing task success from **62% to 88%**, and reducing unsupported conclusions from **18% to 6%**.

AI Engineer – **Intellikgraph (**UC Berkeley Collaboration) Aug 2025 – Dec 2025
- Developed a **knowledge-graph-based maritime reasoning pipeline** integrating vessel trajectories, EEZ boundaries, and maritime events to identify **potentially suspicious maritime activity**.
- Built a **hybrid LLM and symbolic reasoning workflow** that generated SPARQL queries and retrieved supporting evidence from AllegroGraph to verify LLM assessments.

RESEARCH EXPERIENCE

Research Intern – **NSF REU (Computer Vision)** May 2025 – Aug 2025
Advisor: **Dr. Yu Tian**, Assistant Professor, University of Central Florida
- Developed a **training-free multimodal LLM framework** for **video anomaly detection,** combining a **hierarchical reasoning pipeline** with **dynamic memory** for frame- and segment-level context.
- Achieved **81.13%** AUC on UCF-Crime vs. **78.32%** for zero-shot SOTA LAVAD under identical settings.

Independent Research – **Mathematical Foundations of Deep Learning** Jan 2025 – May 2025
Advisor: **Prof. F. Alberto Grünbaum**, Professor Emeritus of Mathematics, UC Berkeley
- Explored deep learning from an applied mathematics perspective, connecting theoretical properties of neural network models and optimization with empirical behavior.
- Designed numerical experiments in Python and MATLAB to analyze training stability and model behavior across controlled hyperparameter and architectural settings.

PROJECTS

Agent Project – **Agent Reliability & Failure Recovery** Aug 2026 – Present
- Built on the **Trial Data agent prototype** with **modular workflows**, **structured validation**, and **failure recovery**; achieved **80%** recovery from injected retrieval and reasoning failures in evaluation.

SKILLS

Languages: **Python**, Java, C++, JavaScript, SQL, SPARQL.
AI/ML: **Agentic AI & RAG**, PyTorch, Transformers, LLM Evaluation.
Frameworks & Tools: **AllegroGraph**, React, Node.js, Git.